

## Week 9

### MCQs (Naïve Bayes classification)

1. What is the primary assumption of the Naïve Bayes classifier regarding feature independence?

- a) Features are completely dependent on each other
- b) Features are independent given the class
- c) Features follow the same probability distribution
- d) Features have equal variance

**Answer:** b) Features are independent given the class

**Explanation:** Naïve Bayes classifier assumes that all features are conditionally independent of each other given the class label. This simplifies computation and makes the model efficient for large datasets.

2. In the context of Naïve Bayes classification, what role does Bayes' theorem play?

- a) It calculates the distance between data points
- b) It determines the number of clusters in the data
- c) It updates prior probabilities with observed evidence
- d) It removes noise from the dataset

**Answer:** c) It updates prior probabilities with observed evidence

**Explanation:** Bayes' theorem is the fundamental principle behind Naïve Bayes classification. It calculates posterior probability by updating the prior probability with the likelihood of observed data.

3. When does the Naïve Bayes classifier perform poorly?

- a) When features are highly correlated
- b) When the dataset is small
- c) When the number of classes is large
- d) When features contain categorical values

**Answer:** a) When features are highly correlated

**Explanation:** The Naïve Bayes classifier assumes that features are independent, which is often not true in real-world scenarios. When features are highly correlated, the classifier may produce inaccurate predictions.

4. How does Laplace smoothing help in Naïve Bayes classification?

- a) It normalizes feature values
- b) It adjusts probability estimates to prevent zero probabilities
- c) It increases the complexity of the model
- d) It reduces computation time

**Answer:** b) It adjusts probability estimates to prevent zero probabilities

**Explanation:** Laplace smoothing (also known as add-one smoothing) ensures that no probability is assigned a value of zero, especially when certain words or features are missing from the training data but appear in test data.

5. Which type of Naïve Bayes classifier is most suitable for classifying binary feature data?

- a) Gaussian Naïve Bayes
- b) Multinomial Naïve Bayes
- c) Bernoulli Naïve Bayes
- d) Decision Tree

**Answer:** c) Bernoulli Naïve Bayes

**Explanation:** Bernoulli Naïve Bayes is best suited for binary (yes/no, 0/1) feature data, such as text classification tasks where presence or absence of words matters instead of their frequency.

6. In Gaussian Naïve Bayes, what assumption is made about feature distributions?

- a) Uniform distribution
- b) Normal distribution
- c) Exponential distribution
- d) Poisson distribution

**Answer:** b) Normal distribution

**Explanation:** In Gaussian Naïve Bayes, the assumption made about feature distributions is that each feature follows a normal (Gaussian) distribution

7. Which Naïve Bayes classifier is best suited for text classification tasks?

- a) Gaussian Naïve Bayes
- b) Multinomial Naïve Bayes
- c) Bernoulli Naïve Bayes
- d) Random Forest

**Answer:** b) Multinomial Naïve Bayes

**Explanation:** For text classification tasks, the most suitable Naïve Bayes classifier is the Multinomial Naïve Bayes (MNB), as it excels at handling discrete data like word frequencies within documents, which is a typical feature of text analysis.

8. Suppose you have a dataset with two classes  $C_1$  and  $C_2$  with prior probabilities  $P(C_1) = 0.6$  and  $P(C_2) = 0.4$ . Given a new observation  $X$ , the likelihoods are  $P(X|C_1) = 0.5$  and  $P(X|C_2) = 0.7$ . What is the posterior probability  $P(C_1|X)$ ?

- a) 0.41

- b) 0.46
- c) 0.51
- d) 0.58

**Answer: c) 0.51**

$$P(C_1|X) = P(C_1) * P(X|C_1) / P(X)$$

$$\text{where } P(X) = P(C_1) * P(X|C_1) + P(C_2) * P(X|C_2)$$

**By putting all the values in the formula, you will get the answer 0.51**

9. What is the main limitation of Naïve Bayes classifiers in Python?

- a) High training time
- b) Poor performance with correlated features
- c) Requires a lot of hyperparameter tuning
- d) Needs large datasets

**Answer: b) Poor performance with correlated features**

**Explanation:** The primary limitation of a Naïve Bayes classifier in Python is its assumption that all features are independent of each other. It performs poorly with correlated features.

10. Which of the following would be the correct classification decision for an instance X if we have three classes with computed posterior probabilities:

$$P(C_1|X) = 0.3, P(C_2|X) = 0.5, P(C_3|X) = 0.2$$

- a) Assign X to  $C_1$
- b) Assign X to  $C_2$
- c) Assign X to  $C_3$
- d) Assign X to random class

**Ans:** Assign X to  $C_2$  as the value of  $P(C_2|X) = 0.5$  which is highest among other two values hence X belongs to  $C_2$  class.